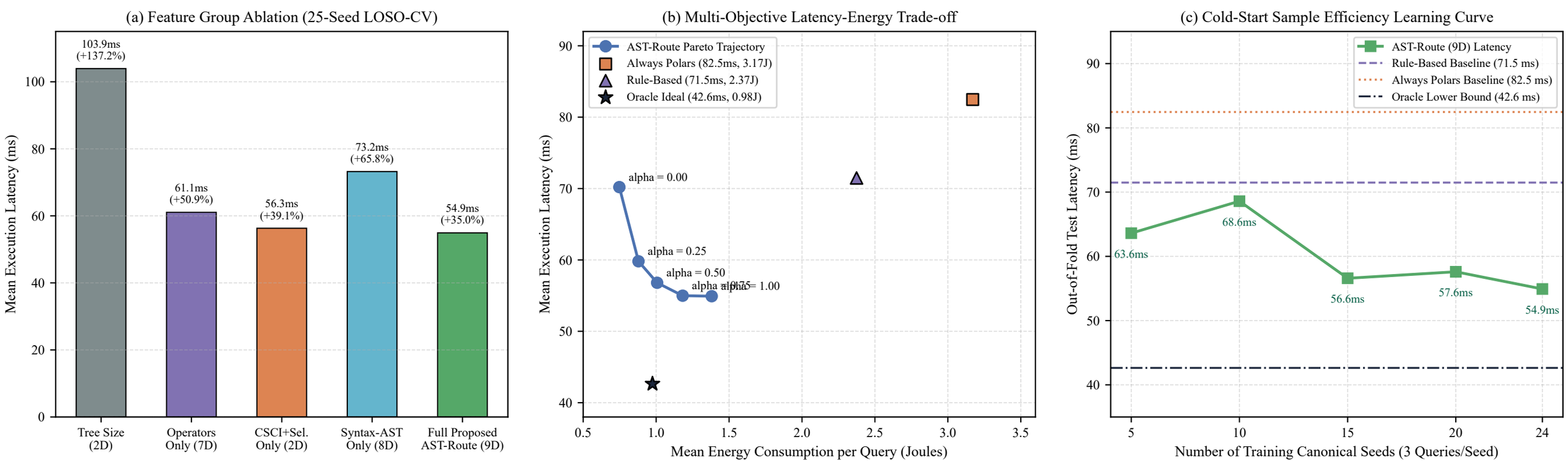
(a) Feature Group Ablation (25-Seed LOSO-CV)
Mean Execution Latency (ms)
103.9ms
(+137.2%)
61.1ms
(+50.9%)
56.3ms
(+39.1%)
73.2ms
(+65.8%)
54.9ms
(+35.0%)
Tree Size
(2D)
Operators
Only (7D)
CSCI+Sel.
Only (2D)
Syntax-AST
Only (8D)
Full Proposed
AST-Route (9D)
(b) Multi-Objective Latency-Energy Trade-off
AST-Route Pareto Trajectory
Always Polars (82.5ms, 3.17J)
Rule-Based (71.5ms, 2.37J)
Oracle Ideal (42.6ms, 0.98J)
alpha = 0.00
alpha = 0.25
alpha = 0.50
alpha = 1.00
Mean Execution Latency (ms)
Mean Energy Consumption per Query (Joules)
(c) Cold-Start Sample Efficiency Learning Curve
AST-Route (9D) Latency
Rule-Based Baseline (71.5 ms)
Always Polars Baseline (82.5 ms)
Oracle Lower Bound (42.6 ms)
63.6ms
68.6ms
56.6ms
57.6ms
54.9ms
Out-of-Fold Test Latency (ms)
Number of Training Canonical Seeds (3 Queries/Seed)